NORC at the University of Chicago

Health Technology Ecosystem

Presented to

Centers for Medicare & Medicaid Services, Assistant Secretary for Technology Policy

Department of Health and Human Services

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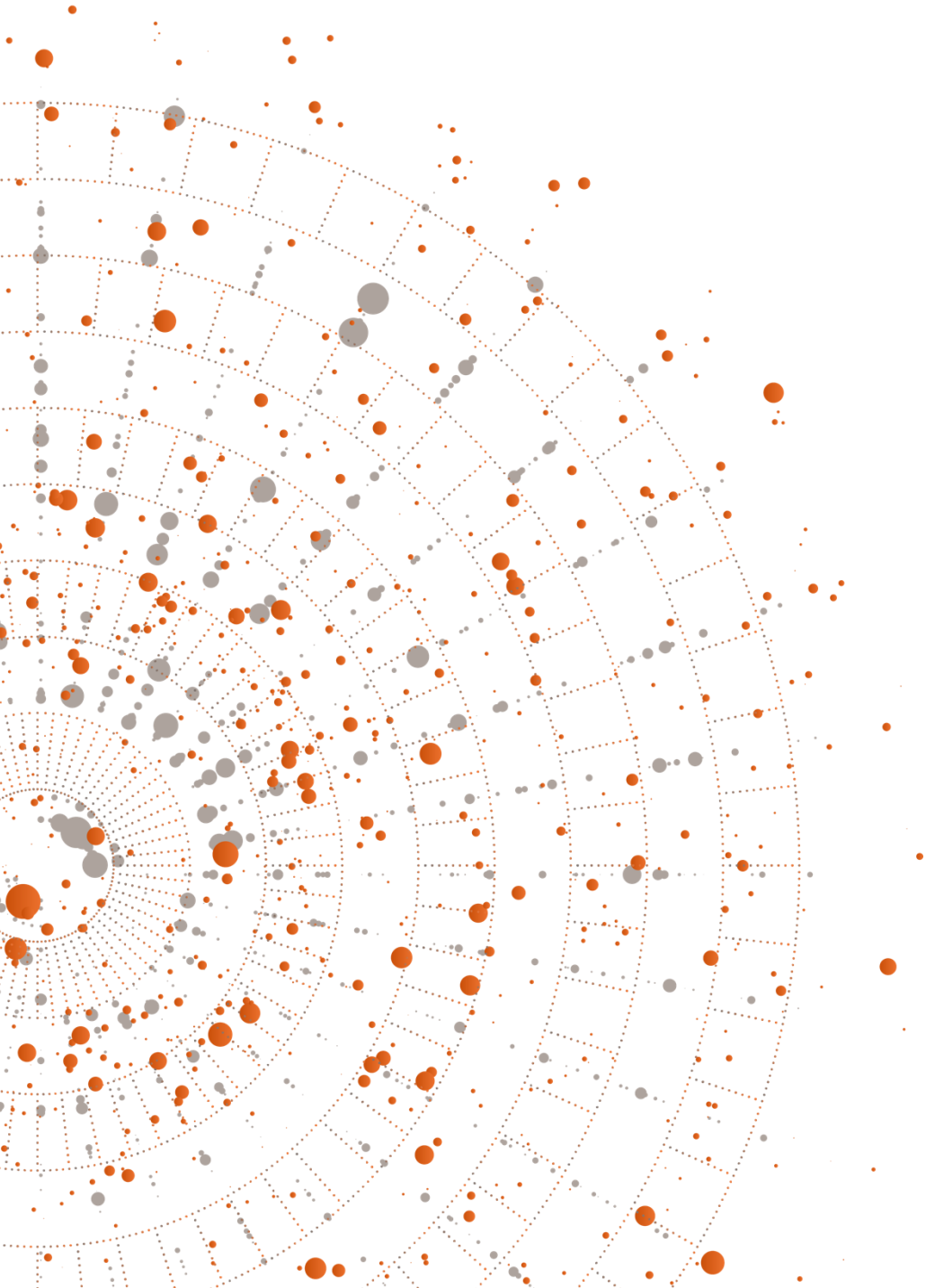

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Table of Contents

Introduction 1

Patients and Caregivers 2

Patient Needs 2

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take? 2

a. What are the top things you would like to be able to do for your or your loved ones' health that can be enabled by digital health products? 4

b. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future. 5

PC-4. What features are missing from apps you use or that you are aware of today? 6

a. What apps should exist but do not yet? Why do you believe they do not exist yet? 6

b. What set of workflows do you believe CMS is uniquely positioned to offer? 7

Data Access and Integration 8

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both? 8

a. What data is valuable but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)? 8

b. What are specific sources, other than claims and clinical data, that would be of highest value, and why? 9

c. What specific opportunities and challenges exist to improve accessibility, interoperability, and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence? 9

Providers 10

Digital Health Apps 11

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients? 11

a. What are the current obstacles? 12

b. What information should providers share with patients when using digital products in the provision of their care? 12

c. What responsibilities do providers have when recommending use of a digital product by a patient? 13

Conclusion 14

Introduction

NORC at the University of Chicago (NORC) is pleased to respond to the Request for Information (RFI) from the Centers for Medicare & Medicaid Services (CMS) and the Assistant Secretary for Technology Policy (ASTP/ONC). We welcome the opportunity to provide our perspective on digital health products for Medicare beneficiaries, as well as the state of data interoperability and the broader health technology ecosystem.

Established in 1941, NORC is one of the largest and most respected nonpartisan, nonprofit research organizations in the United States and remains at the forefront of conducting rigorous research for the public's interest. NORC has a longstanding history of supporting federal initiatives focused on improving accessibility and quality of health information. In the last two decades, our work has expanded to include a growing emphasis on digital health, particularly in efforts to improve digital health products available to patients, including Medicare beneficiaries, to support informed decision-making, better health outcomes, increased healthcare efficiency, and reduced costs. NORC's work has also focused on expanding access to timely and accurate health data for all stakeholders—including clinicians, researchers, and policymakers—to facilitate improved patient care and population health outcomes.

NORC has collaborated extensively with provider organizations, patient advocacy organizations, electronic health record (EHR) developers, and health informatics organizations to conduct research that aims to drive better digital health products and advance the digital health ecosystem for care teams, patients, and caregivers.[1] [2] [3] [4] [5]Additionally, NORC has partnered with the Department of Health and Human Services (HHS) offices[6] [7] and other federal agencies on these priorities, including CMS and ASTP/ONC. NORC has also worked with the Patient-Centered Outcomes Research Institute (PCORI) to advance the science of engaging patients in research[8] regarding their needs and preferences and expand stakeholder access to health data. As part of these initiatives, NORC's work has included:

- Designing and developing digital technologies to better engage patients in their care.
- Enhancing the data infrastructure needed to conduct meaningful patient-centered outcomes research, deliver evidence-based patient-centered care, and improve healthcare efficiency.
- Engaging directly with patients to explore preferences and concerns around collecting and using patient-contributed data in healthcare and research settings.

NORC maintains a commitment to prioritizing patients' perspectives and experiences throughout this work. To that end, NORC actively engages patients to better understand their perspectives, preferences, and experiences, ensuring that the design, development, and implementation of digital health tools in healthcare are truly patient-centered and grounded in the realities of those most impacted by these technologies.

The following responses reflect NORC's insights on digital health products for Medicare beneficiaries, data interoperability, and the broader health technology infrastructure, informed by our research in collaboration with a diverse range of stakeholders, including patients and caregivers, healthcare providers, health systems, informaticians, and app developers. NORC recognizes the importance of increasing beneficiary access to digital health tools, which is essential for empowering beneficiaries in their own care decisions and for enhancing the quality and scope of health research through improved data availability and system interoperability. Furthermore, digital health products can reduce waste in

healthcare costs for the government by improving efficiency, supporting preventive care, and facilitating chronic disease management.[9] [10] [11]

Patients and Caregivers

Providing patients with the right information at the right time and in the right format and channel can help empower them to make evidence-informed decisions about their health care, which can lead to better outcomes.[12] Health management applications can make it easier for patients to submit and access their health information, allowing them to feel empowered in discussions with clinicians about which care options will be best for them. For example, clinical decision support tools can collect and analyze patient-reported blood pressure data to support patients and their care teams in making the best decisions for an individual's hypertension management based on their circumstances and preferences.[13] Patients have also shown an increased desire to access their health information through personal health record apps. For example, 57 percent of patients reported accessing at least some of their medical data via the patient portal in 2022, representing a nearly 50 percent increase since 2020.[14] When patients can make decisions that align with their preferences, needs, values, and specific circumstances, it can lead to improved health outcomes, increased satisfaction, and greater active engagement in their healthcare journey.[15]

The rapid advancement of digital health products and supportive policies towards a more patient-centered healthcare delivery model creates a ripe environment for advancing the accessibility and quality of health information for patients.[16] For one, there is an increased availability of patient-specific information that can inform digital health products, including patient-reported outcomes (PROs) data[17] [18] and patient preference data.[5] Additionally, there has been a proliferation of digital health products, including health management applications, personal health records apps, and wearable devices, that allow patients to be more engaged in their healthcare by providing them with real-time medical data as well as information on their goals and values related to health and wellness.[19] [20] Advancements in healthcare interoperability, as well as recent requirements for developers to enable patient access to their medical data, make it easier for patients to access and use information to manage their health conditions in partnership with their clinicians.[21] Finally, rapid developments in artificial intelligence (AI)[22] technologies, such as conversational agents that can facilitate communication between patients and clinicians,[23] create opportunities to analyze vast amounts of data to support more personalized care.

Patient Needs

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

Through NORC's engagement with patients and patient advocates in qualitative interviews, roundtable discussions, and focus groups regarding health management apps for patients, we have gained valuable insights into what patients expect and need from digital health products. These discussions have provided a nuanced understanding of how patients interact with health technologies and what features make these tools useful and trustworthy. Several themes have emerged about what is most important to patients in health management and care navigation apps.

- **Apps That Support Health Tracking and Care Navigation.** Patients find value in apps that enable them to have an active role in managing their health.[24] Patients increasingly use apps that support self-management by allowing them to track symptoms, medications, appointments, and progress

toward health goals. Patients also use these apps to share health information, such as patient-reported outcomes, with their clinicians.[25] Patient-facing apps that leverage AI algorithms may help patients detect patterns and trends in their symptoms and health data points, which gives them even more information about adjusting their lifestyle to manage their health. These apps are particularly useful for patients managing multiple chronic conditions as they can help coordinate care across multiple clinical settings and data tracking related to their conditions.

- **Apps That Provide Easy, Organized Access to Personal Health Information.** Patients have emphasized the importance of apps that facilitate timely access to their health information, including test results, medication lists, and care plans. This access enables them to make informed decisions with their providers.[4] Patients have also indicated the importance of apps presenting this information in an accessible manner that is intuitive and clear.[26] For many patients, especially those managing multiple conditions, quick and consistent access to their health information can be critical for coordinating care. [27]
- **Apps That Facilitate Communication with Clinicians and Care Teams.** According to patients, a major value of health apps is their ability to bridge communication gaps between patients and their clinicians.[28] NORC's research with patients has emphasized that they appreciate apps that allow them to send messages to their clinicians, receive updates or test results, and ask follow-up questions between appointments.[29] Many patients have also shared that they support features that enable real-time health data sharing—such as outcomes from remote monitoring devices or patient-reported outcomes (PROs)—to enable clinical decision-making. There is growing interest in using conversational agents or chatbots within these apps to make interactions with care teams more seamless and responsive.[30]
- **Apps That Give Patients Control Over Their Health Data and Access.** NORC's qualitative research with patients indicates that they want to control how their health information is used, and who has access to it.[4] They desire digital health products that allow them to upload or input their data—such as symptoms, blood pressure readings, or mood diaries—and to share that data with their clinical team or researchers as they see fit.[29] Equally important is the ability to grant access to this data to caregivers, family members, or other trusted individuals.[31] This is especially relevant for older adults, people with disabilities, or patients managing complex care who may rely on support from others to organize or interpret their health information.

Patients have also noted important design considerations of health apps that can significantly impact their usability and effectiveness. Apps should be:[25] [32]

- Easy to use with intuitive interfaces, requiring minimal steps to complete tasks; navigation of the apps should accommodate varying levels of digital literacy.
- Understandable with content presented in plain language.
- Accessible with features that include people with visual, auditory, cognitive, or other impairments and compatible with assistive technologies.
- Respectful of preferences, including preferences for notifications and display features.
- Transparent data use and use of AI in digital health products with clear information about how patient data is collected, used, stored, and shared.
- Protective of privacy by providing assurances that sensitive health information is secure, and that consent is sought before secondary uses of patient data.

These insights reinforce the importance of designing health management apps that align with patients' values, needs, and real-world experiences.

a. What are the top things you would like to be able to do for your or your loved ones' health that can be enabled by digital health products?

In our qualitative research, patients and patient advocates identified several key actions they would like to be able to take for their own or their loved ones' health—actions they believe can be supported by digital health products. These include sharing patient preferences, which speak to how they want to engage with their care with clinicians, coordinating care across multiple providers and settings, managing chronic conditions, supporting medication adherence, and supporting caregivers in care navigation.[26 31 33]

- **Enable Patients to Share Preferences.** NORC's qualitative research has demonstrated that patients want to easily share their preferences and priorities with clinicians, which can be enabled by digital health products. This includes indicating goals of care, preferred communication styles, values related to treatment decisions, and advanced care planning choices. This functionality is especially important for those navigating serious or chronic illnesses, where patient preferences, values, and quality-of-life considerations may strongly influence treatment options.[31]
- **Coordinate Care.** Patients often manage care across multiple clinicians and care settings, which can lead to fragmented communication.[34] In our discussions with patients, they have expressed a strong interest in digital health products that help bridge these gaps by centralizing key health information, sharing visit summaries across care teams, enabling remote monitoring, and providing visibility into care plans or upcoming appointments.[33] Digital health products that include care coordination features can help patients avoid duplicate testing, ensure consistent messaging across providers, and reduce the burden of relaying the same information multiple times.[35]
- **Support with Chronic Disease Management.** Research indicates that patients living with conditions such as diabetes, heart disease, asthma, or arthritis want to be able to better manage their conditions daily.[36] Digital health products can support chronic disease management through monitoring symptoms, logging daily activities, visualizing trends in health data, and receiving real-time guidance based on their inputs. These features not only promote better self-management but also facilitate more informed and productive conversations with healthcare providers.[37]
- **Support with Medication Adherence.** Research indicates that despite motivation, patients face barriers with medication adherence.[38] Patients have expressed interest in digital health products that can provide timely reminders, track medication schedules, offer education on drug interactions or side effects, and flag issues that warrant follow-up with a clinician.[39]
- **Enable Caregivers to Navigate Care Decisions.** Caregivers have an important role in managing their loved one's health. Research indicates that caregivers want to access timely health information to participate in care decisions alongside their loved ones and their care teams.[40 41]Digital health products can support this by enabling shared access to care plans, medication lists, clinical notes, and appointment schedules. Products that offer educational resources, communication channels with providers, and decision aids can further empower caregivers to make informed decisions and advocate on behalf of their loved ones. [42 43] This is especially important for patients with limited capacity to manage their care due to age, disability, or illness.

These functions within digital health products can empower patients and their caregivers to make evidence-informed decisions about their healthcare, leading to better healthcare outcomes[12] and reducing inefficiencies in healthcare delivery.[11]

b. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future.

Through qualitative interviews with patients, NORC's research has identified a range of needs that could be supported by a digital health product functioning as a personal assistant (i.e., software solutions potentially supported by AI). This includes support with tasks that make managing health easier, more efficient, and more personalized. Patients and caregivers also shared that they value solutions that can help them make informed decisions and navigate care complexities. Below, we emphasize these tasks and how they could be facilitated by future software solutions such as generative AI.

- **Generate Plain Language Summaries.** Research demonstrates that patients appreciate plain language summaries of clinical documents such as lab results, imaging reports, or clinician notes.[44] Generative AI can serve as a personal health assistant, translating medical jargon into plain language, thus empowering patients to understand their health status and care plans.[45]
- **Supplement Patient Portal Communication.** NORC's report on patient preferences emphasizes that patients want to communicate seamlessly with their providers between clinical visits as needed.[31] AI-powered chatbots can provide timely, conversational responses and escalate more complex queries to clinicians when needed.[46]
- **Provide Tailored Health Recommendations**. Generative AI can conduct conversational assessments, such as nutrition coaching, where it can probe for and identify patients' preferences, dietary habits, barriers, and resource constraints to provide personalized nutrition plans and meal suggestions. Over time, it can provide recommendations based on feedback and outcomes, mimicking the role of an empathetic, context-aware health coach.[47]
- **Monitor Health Conditions**. Generative AI can monitor health conditions such as asthma and chronic obstructive pulmonary disease from smart inhalers and environmental sensors to collect real-time data on medication usage and environmental factors, including air quality. The AI can generate plain language insights and generate recommendations or alerts for patients and clinicians.[48]
- **Manage Disease**s. Generative AI can integrate data from various sources to manage diseases such as diabetes. It can analyze continuous glucose monitoring data, dietary habits, and physical activity to generate actionable alerts for users.[48]
- **Support Medication Adherence**. Generative AI can track medication intake via smart pill bottles or visual confirmation (e.g., smartphone camera) to remind patients, flag missed doses, and adjust communication strategies based on adherence patterns.[49]

Agentic AI is an emerging type of AI that acts autonomously, with minimal human supervision, to pursue specific goals, make decisions, and take action on behalf of users.[50] In healthcare, agentic AI can be utilized to fulfill the role of a personal health assistant, supporting patients' health needs regarding care coordination, treatment adherence, and making informed decisions. Examples of agentic AI that may be useful for assisting patients include personalized health monitoring (e.g., integrating data from wearable devices and providing real-time feedback and alerts), dynamic treatment planning (e.g., analyzing patient data to suggest adjustments in treatment plans), proactive disease management (e.g., identifying early signs of disease exacerbation and recommending preventive measures or interventions), and enhanced patient engagement (e.g., facilitating communication between patients and clinicians, and simulating empathetic and context-aware back-and-forth communication with providers).[51]

While there are promising areas for AI use in healthcare, specifically as it pertains to supporting patients with their healthcare needs, there are important potential risks associated with the continued and further adoption of AI systems. These potential risks include ethical concerns, [52] hallucinations (or

confabulations) and other technical issues,[53 54] liability of AI algorithms,[55] workforce considerations,[56] transparency,[57] bias,[58 59] and patient safety.[60] Furthermore, there may be issues with trust in AI among patients as they have reported mixed views on its use in their healthcare. While many patients believe that AI has the potential to strengthen healthcare delivery, [61] there is a need to balance the opportunities of AI use in healthcare with the need to manage risks and maintain trust.

PC-4. What features are missing from apps you use or that you are aware of today?

Through roundtables and qualitative interviews with patients and other stakeholders, literature reviews, and pilot testing of tools, NORC's research has identified gaps in current health management applications that limit patient involvement in care planning. These gaps center on the following themes:

- **Ability to Submit Patient Preferences.** Collecting data on patient preferences can ensure that care is aligned with patient needs and values.[62] NORC's research found that data within some types of patient preference domains were more regularly collected and ready for standardization and use in health management applications compared to others. For example, there are standards and codes available to capture preferences for personal characteristics (i.e., preferred name, language), healthcare services (i.e., treatment, advance care directives), and engagement (i.e., receipt of information, inclusion of others in decision making), but limited standards to capture preferences related to care experience and how patients want to access their data.[63] Additionally, health management applications should allow patients to update their preferences, as patients' needs and preferences can change over time. Functionality within patient apps that facilitates dynamic collection of patient preference data is needed to ensure fluidity of preferences over time.
- **Enhanced Communication.** Patients and other stakeholders have agreed on the potential for health management applications, particularly those that leverage AI, to streamline patient-clinician communication. While some patients may prioritize face-to-face interactions with their provider, others may be comfortable substituting asynchronous communication through an app for interactions.[61] Even with app-mediated asynchronous communication, patients desire a more natural, flexible form of communication that simulates a back-and-forth conversation.[64] Findings from NORC's research indicate an opportunity for AI-supported health management apps, particularly chatbots powered by generative AI, to fill this gap. Given the volume of back-and-forth messaging between patients and providers, Generative AI-powered health management applications can be used to communicate with patients to clarify their needs and symptoms between visits, enabling more efficient communication with providers during visits.[32] However, appropriate guardrails must be put in place when using this type of technology to protect patient privacy, ensure safety via clinician involvement, as assuage liability concerns from the clinician side.[65]
- **Feedback Loops.** Patients submit large amounts of data through health management applications, including symptom information, biometric information, information about goals, and medication tracking information.[29] The results from pilot testing of an AI-enabled health management application found that patients desire feedback loops with their clinicians; they want to know when the information they submit through the app has been received by a clinician so that it can be used in future decision-making or receive a response from their clinician about a data point they submitted.[26] However, these feedback loops are largely missing from apps.

a. What apps should exist but do not yet? Why do you believe they do not exist yet?

NORC's research has reviewed the current landscape of digital health app interoperability and identified the following gaps in the field:

- **Apps that can Communicate across Health Systems.** To fully realize the potential of patient-facing health management applications, they must be interoperable within and across the health IT systems that care teams use, particularly EHRs. A lack of interoperability within apps causes a burden to patients, as it can contribute to incomplete medical records, which necessitate manual conveying of information to clinicians.[66] One of the leading standards that can support health management application interoperability is Health Level Seven (HL7®) Fast Healthcare Interoperability Resources (FHIR®), which allows different systems to efficiently exchange healthcare data in a consistent format.[67] However, despite the increasing adoption of EHR and app developers, FHIR standards and resources are not being used ubiquitously – fewer than a quarter of healthcare organizations report adopting FHIR application programming interfaces (APIs) at scale.[68] Additionally, the sharing and use of patient app data across care settings is impeded by insufficient data standards (development and adoption) and differences in semantics used across tools and systems.[69] Greater coordination is needed across health management application developers and EHR developers to ensure interoperability between apps and EHRs and allow patient data collected by apps to be integrated into clinician workflows across the broader health IT ecosystem.
- **Apps that Address Multiple Conditions.** Findings from the research have shown that health management applications are increasingly being developed for specific populations or specific uses, such as hypertension, diabetes, mental health conditions, or medication adherence.[70] [71] A current gap in the app marketplace is apps that address multiple conditions at once using a shared interface. The limited availability of multiple condition apps leads to additional burdens for patients who have to routinely use numerous apps to address their conditions.[72] Apps that address multimorbidity would be especially important for the more than 25 percent of the US population with multiple chronic conditions and whose complex care needs are addressed across multiple providers and care settings.[73] Work is currently being done to develop and test shared electronic care planning apps to enable more effective communication and coordination about a patient's treatment plan, goals, and preferences across care settings.[74] However, these tools are not the norm due to challenges with the availability and functionality of FHIR-based interoperability systems across care settings, as discussed above.

b. What set of workflows do you believe CMS is uniquely positioned to offer?

Through digital health products, CMS can leverage its quality measurement and reporting initiatives to promote more efficient tracking and assessment of patient health outcomes, including PROs. The CMS Measures Management System (MMS)[75] is a standardized source of quality measures that can support the development of relevant, patient-centered measures for digital health products across the Measure Lifecycle (from specification to measure use), which is currently a gap in the field.[13] The CMS Meaningful Measures 2.0 can also guide digital health products in collecting and analyzing only the most impactful data, making them more user-friendly for patients and caregivers.[76]

CMS is uniquely positioned to improve the integration of patient preferences into health management applications, which are central to patient-centered care. Despite an increased focus on delivering patient-centered care that empowers patients to participate in care decision-making, few health management applications consider patient preferences when offering clinical recommendations and guidance to support shared decision-making.[77] CMS can support the standardization of these preferences to facilitate their integration into apps. For example, by requesting new terms that represent patient preferences from existing data standards organizations, such as Logical Observation Identifiers Names and Codes (LOINC)[78] and Systematized Nomenclature of Medicine (SNOMED),[79] or raising awareness for the inclusion of additional patient preference codes in future versions of the United States Core Data for Interoperability (USCDI). USCDI V4 includes patient preferences related to preferred method and language of communication, advance directives, care experience preferences, and goals and health concerns, but there is limited adoption of LOINC codes capturing preferences related to treatment

intervention and care experience.[80] Ultimately, health management applications will be more impactful and meaningful to patients when they account for patient needs and preferences.

CMS is also well-positioned to help identify high-priority clinical areas, aligned with broader HHS goals, where health management applications would be most effective at improving the quality of care. CMS operates the Quality Improvement Organization (QIO) Program[81] and other quality improvement initiatives that focus on the nation's most pressing health topics, such as chronic disease management, behavioral healthcare coordination, and long-term services and supports.[82] Using evidence from these programs, CMS can provide guidance on areas where digital health will have the greatest impact.

Data Access and Integration

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

In NORC's research with patients they have consistently emphasized the importance of sharing their personal preferences, priorities, and values with their care teams. While other forms of data are readily available through patient portals and EHRs, such as demographic or clinical data, patients are interested in ensuring their preferences are readily available and acted upon in their care plans.[3] Specifically, patients have expressed that their preferences pertaining to the nature of their engagement (e.g., the degree to which a patient wants to be engaged in healthcare decisions, how they want others engaged, mode of engagement), access to information (e.g., what information they would like to access), communication (e.g., preferred mode and frequency of communication), caregiving (e.g., caregiving needs and how/what information is shared with caregivers), and treatment (e.g., palliative care, aggressive treatment preferences, and advance directives) are high priority data they would like their clinicians to know and to be shared across healthcare teams.[31]

a. What data is valuable but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?

Claims Data. An important data source that is valuable but difficult to access is claims data, including Medicare, Medicaid, and commercial claims data. These data are valuable in that they provide a longitudinal view of a patient's healthcare utilization across various settings[83] (e.g., outpatient, inpatient, pharmacy), offering insight into treatment patterns, chronic conditions, and trends through diagnostic and procedure codes. If delivered in a usable way, these data can be valuable to patients, caregivers, and other stakeholders to detect gaps in care (e.g., inconsistent medication fills) to tailor care coordination and management accordingly. The challenges with claims data include limited access to patients and caregivers for their own use, restricted access to developers or vendors due to privacy regulations, and the fact that claims data is retrospective, which limits its usefulness for real-time applications. CMS's Blue Button 2.0 initiative laid the initial groundwork for enabling API-based patient access to Medicare claims data, but it is not widely used.[84]

Clinical Notes. Clinical notes are another important data element that captures context, nuance, and patient-provider discussions not found in structured fields. These data can provide insights into patient preferences, social needs that impact health outcomes, and other contexts that can enable a more holistic, patient-centered approach to healthcare and care planning. Clinical notes are often not shared with patients or shared inconsistently across systems. Additionally, unstructured text makes it challenging for technical vendors to extract actionable information.[85] OpenNotes adoption by EHR systems has promoted patient access to clinical notes.[86]

Pricing/Cost Information. Additionally, pricing information is valuable to patients and caregivers as it further enables them to make informed decisions regarding their healthcare.[87] Patients want to know expected out-of-pocket costs for their healthcare and medications, but pricing data is often missing or inconsistent. Additionally, app developers may struggle to access real-time or personalized cost estimates due to the complexity of payer systems.

b. What are specific sources, other than claims and clinical data, that would be of highest value, and why?

In addition to claims and clinical data, patient-contributed data, such as patient preferences, PROs, and nonclinical contextual data, are among the most valuable yet underutilized sources of information for improving healthcare and facilitating patient empowerment. The use of patient preference data helps ensure care aligns with patients' preferences, which can lead to more active participation in their care, higher satisfaction, and treatment adherence.[88] PROs, often collected through apps or web-based surveys, are information from the patient regarding their health status, symptoms, care experience, outcomes, and more without clinician judgment or interpretation. PROs can provide insight from the patient that complements clinical information and helps tailor treatment plans, ultimately improving patient satisfaction.[89]

Nonclinical contextual information collected directly from patients or obtained from other data systems, such as social services systems, provides additional valuable insight regarding patients' social needs and barriers. Research demonstrates that nonclinical factors, such as social and environmental factors, account for up to 80 percent of health outcomes.[90] Without the context of these factors, providers may miss key drivers of a patient's clinical condition or factors that may impact their ability to comply with treatment plans. Integrating these nonclinical data enables more personalized care for patients and allows clinicians and care teams to tailor treatment and address barriers to care (such as transportation or food insecurity). Typically, this nonclinical information is collected from patients through questionnaires during clinical visits or through patient portals or entered into structure fields or clinical notes in the EHR.[91]

c. What specific opportunities and challenges exist to improve accessibility, interoperability, and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?

Multiple challenges and opportunities exist regarding data accessibility, interoperability, and integration from different sources. In addition to enabling more meaningful clinical research and actionable evidence, opportunities to enhance clinical data accessibility, interoperability, and integration challenges can also reduce unnecessary healthcare inefficiencies and spending.[92] [93] Based on NORC's research, we highlight several in the table below.[29] [65] [94]

Challenges	Opportunities
Variability in EHR Systems and Standards EHR systems vary widely in how they capture, store, and structure data. This heterogeneity complicates efforts to enhance accessibility, interoperability, and integration of clinical data from different sources.	**Advance Interoperability Standards** Standards such as HL7 FHIR (Fast Healthcare Interoperability Resources®) and USCDI (United States Core Data for Interoperability) offer a foundation for consistent data exchange. App developers can use these standards to help harmonize data formats and improve interoperability across systems.
Limited Application Programming Interfaces (APIs)	**Use Patient-Facing APIs**

Challenges	Opportunities
There are limited robust APIs within health systems that allow third-party app developers to access and exchange data securely.	App developers can develop FHIR-based patient-facing APIs that enable patients' secure communication with their care team.
Varying Levels of Standards Maturity and Adoption The adoption and implementation of standards such as HL7 FHIR and SNOMED CT are uneven across health systems. While some organizations have embraced these standards, others rely on legacy or proprietary formats, creating gaps in data interoperability.	**Federal Initiatives** Federal initiatives like the *21st Century Cures Act* drive regulatory and technical frameworks to promote data sharing and patient access.
Variability in Data Type Data from various sources—such as EHRs, wearable devices, and lab systems—often differ in format, structure, and terminology, making it challenging to integrate them in a way that supports meaningful analysis and decision-making.	**Use AI to Support Data Harmonization and Standardization** Developers can use AI to standardize formats, map terminologies, and harmonize disparate sources using machine learning and natural language processing.
Variability in Data Quality Missing values, inconsistent entries, and measurement errors can compromise the reliability of integrated health data, making it more difficult to draw accurate insights or support clinical decisions from multiple data sources.	**Use AI to Support Data Imputation and Quality Improvement** Developers can use AI to detect missing or inconsistent data and use statistical modeling (such as data imputation) to fill gaps or flag anomalies.

Providers

Patient-focused digital health products can supply providers with the information and evidence needed to enhance shared decision-making and ultimately improve patient outcomes. Tools such as patient portals and telehealth platforms can enhance communication between providers, patients, and caregivers, which can influence patient self-management and patient-centered outcomes.[95] Clinical decision support tools integrated within EHRs can also streamline workflows and reduce administrative burden for clinicians by automating tasks such as test result notifications. Remote monitoring devices and mobile health apps can facilitate access to real-time information outside of clinic settings, which allows providers to make timely adjustments to care plans between patient visits[96] and can reduce healthcare costs. [97]

Recent advancements have promoted the adoption and effective use of digital health products by providers and healthcare systems. Federal initiatives such as the *21st Century Cures Act* and its associated ONC's Cures Act Final rule support the seamless and secure access of health information through EHRs, improving access and availability of health data and encouraging interoperability between healthcare systems.[98] Interoperability standards such as FHIR APIs enable this seamless exchange, making it easier for clinicians to access patient data within EHRs and their regular workflows.[99] Finally, AI has been increasingly incorporated into healthcare settings and has the potential to accelerate opportunities for patient-centered care and efficiencies in care delivery.[100] With these advancements, there is an increasing

need to ensure modern digital health products are designed safely and with clinician and patient needs in mind.

Digital Health Apps

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?

CMS and its partners can consider conducting assessments and/or systematically collecting information that could be used to demonstrate the value of digital health products for clinicians. In particular, these assessments can focus on demonstrating how digital health products reduce burden or improve efficiency within clinical workflows. This is especially important for overburdened clinicians who could leverage digital health products to streamline care and free up more time to focus on patient interaction and safety.[65] In addition, these types of assessments can unpack whether and how digital health tools impact patients' access to health information and evidence-based care recommendations. This is valuable for clinicians serving patients in rural areas, where improved asynchronous communication would reduce patients' need to travel long distances to receive healthcare.

Limitations in broadband access, particularly in rural areas, have implications for the use of digital health products. Even though smartphone and computer use are nearly ubiquitous in the United States,[101] some patients in low-income or rural areas do not use health management applications because they do not have regular internet access to their computer or smart phone.[102] [103] [104] To address this challenge and support providers in facilitating access to care for patients, CMS and its partners can prioritize evaluation studies of text messaging-based health management applications. Text message-facilitated health management interventions can provide patient support, deliver reminders or nudges, and collect data directly from patients in an efficient, low-cost manner.[105] Text message notifications are associated with greater patient engagement and have the ability to reach more patients across a variety of clinical and socioeconomic contexts.[106] CMS and its partners can conduct evaluations to ensure that text messaging-based interventions work on basic mobile phones to support access for populations who may not own a smartphone or have limited internet access.

Health management applications have the potential to improve healthcare delivery by ensuring that patient-generated health data, patient-reported outcomes (PROs) data, and patient preferences data are adequately captured and communicated to clinicians. Through rapid cycle evaluations, CMS can contribute to demonstrating the value of health management applications and other digital health products in facilitating information processing, supporting bidirectional communication between patients and providers, and capturing important information outside of routine health visits.

Rapid cycle randomized control trials (RCTs) can be used to quickly identify which digital health tools work and do not work in order to quickly improve healthcare practice. Researchers have used these techniques to test and optimize digital health tools for scheduling reminders, prescription reminders, vaccine alerts, and personalized messages, among others.[107] CMS can consider funding these rapid-cycle RCT projects to act as accelerated quality improvement projects, wherein digital health products can be tested, fine-tuned, and adapted for real-world use based on end-user feedback. Findings from the RCTs can be used to illustrate to clinicians how digital health products will improve their workflows and meet the needs of their patients.

a. What are the current obstacles?

The introduction of digital health products into clinical care requires integrating data from disparate sources into the EHR and sustained patient engagement. If these solutions are not implemented correctly with the appropriate workflows, clinicians' cognitive burden could increase.

EHR and app developers have faced significant challenges integrating patient-generated data from digital products into existing EHR database structures.[108] These challenges include lack of agreement on the format in which data should be collected from apps (e.g., raw data versus summaries), where patient-generated data should be stored in the EHR, which standards should be used to encode the data, and reticence to integrate patient-generated data with clinician-collected data due to concerns about protecting EHR data integrity.[109]

Digital health products that rely on repeat physiologic measurements (e.g., for diabetes or high blood pressure management) or behavior tracking (e.g., medication adherence) require sustained patient engagement in submitting data. However, app developers and health systems face challenges keeping patients engaged with digital health products, as their use of these apps typically drops after a few months.[108] Few studies have examined the factors that improve patient engagement.[13] Still, potential reasons include the narrow focus of apps on specific conditions and the preference among patients to use their patient portal as a more comprehensive source of information rather than submitting and viewing data on a separate device or app.[108]

Digital health products, while an important source of additional patient-generated health data can potentially increase clinician burnout and fatigue by adding another source of information for clinicians to review. The siloed nature of digital health products, which are often disease-specific rather than targeting multiple conditions at once, can contribute to workflow fragmentation and increased inefficiency for clinicians. Clinicians would now be responsible for reviewing patient-contributed data from apps, which can be viewed as extra work in the decision-making process.[110] This can lead to increased cognitive burden on care teams and contribute to fatigue.[111] Technology that addresses alert fatigue leverages AI and clinical decision support system optimization to improve the relevance of alerts clinicians receive, reduce alert volume, and enhance the user experience, making it easier for clinicians to navigate digital health products.[112, 113]

b. What information should providers share with patients when using digital products in the provision of their care?

Trust is fundamental to successfully using health management apps and other digital products. Source credibility – or the degree to which a message recipient perceives a message and/or its sender as credible[114] – is critical to building trust and mediates the relationship between health communications and uptake of guidance.[115] Findings from the research highlight strategies developers, clinicians, and other stakeholders can adopt to improve source credibility and digital products.[116] These strategies include:

- **Communicating key information regarding the construction and creation of health management apps.** Providing patients with information about the embedded guidelines, data, and algorithms used in health management apps as well as their intended uses and limitations increases transparency.[117]
- **Tailoring messaging**. Clinicians can inform patients whether a digital product's delivery and format are tailored to end users' needs and characteristics. Patients prefer tailored information and recommendations whenever possible.[118]

- **Creating feedback loops.** Clinicians can communicate to patients the extent to which digital tools enable bidirectional exchange. The provision of feedback loops signals openness and receptivity by creating streamlined mechanisms for both patients and clinicians to share their feedback.[119]
- **Promoting transparency.** Providers should be transparent about why and how data being collected from patients through digital health tools are being used by clinicians.[120] This explanation may make patients more willing to share information before and during healthcare visits.

Additionally, patients interviewed in NORC's research emphasized the need for transparency regarding the use of AI-enabled digital health products in their care. Patients identified several factors related to transparency that would influence their level of trust in AI-powered digital tools that should be considered:

- **Identifying when AI is used in care.** Patients want to be informed when an AI tool is being used in their care, and they want these tools to be accessible, easy to understand, and friendly. Clinicians should explicitly inform patients when a digital product they are recommended leverages AI, and give patients the ability to opt out, if desired.
- **Understanding access to patient data.** Patients emphasized that digital products should provide transparent information on data ownership, making it clear who can access the tool's data and with whom it is shared. In order to build trust, digital products should provide adequate information how patients can access, update, or delete their data.
- **Receiving information on underlying data sources.** Patients agree that they would trust AI-enabled digital products more if they knew it was providing accurate information. To the extent possible, explainable AI tools should be used, as they provide clear justifications for the recommendations they make.[121]

c. What responsibilities do providers have when recommending use of a digital product by a patient?

The proliferation of digital health products has facilitated the collection of patient-generated data outside of healthcare visits. Additional supports are needed within healthcare systems to help clinicians manage the additional patient-generated data introduced by these tools.

In order to promote the responsible use of digital health products by patients, new policies, procedures, and standards may be necessary to 1) ensure patient-generated data are appropriately reviewed and managed between visits, 2) facilitate incorporation of patient-generated data into existing EHRs, and 3) troubleshoot potentially erroneous data from digital products.[108]

Additionally, new or revised reimbursement models may be necessary to support the collection, management, and use of patient-generated data from digital health products in clinical workflows. These reimbursement policies would be important to account for the additional receipt of data outside of typical clinic work hours, as healthcare organizations would otherwise be forced to internally fund these activities.[108]

Finally, providers can consider leveraging digital health formularies to streamline the process of recommending and tracking digital health products for patients. Digital health formularies are curated lists of digital health products, such as health management applications or care navigation applications that health organizations have selected and vetted for end users.[122] Digital health formularies make it easier for providers to recommend digital health products that are safe and effective, which is especially important given that patients are more likely to use apps that have been recommended by healthcare professionals.[123]

Conclusion

We appreciate the opportunity to respond to the RFI and to provide our perspective on the current state of digital health products and data interoperability. Increasing Medicare beneficiary access to digital health tools that meet their needs can empower them in their healthcare decisions, improve health outcomes, and reduce government healthcare costs by enhancing healthcare efficiency. We hope that our responses support CMS and ASTP/ONC in improving the health technology ecosystem in a patient-centered way.

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